Homogeneous differential equation: The differential equation $M(x,y)dx + N(x,y)dy = 0$ is called n dimensional homogeneous differential equation if and only if

$$M(x,y) = x^n M_1\left(\frac{y}{x}\right), \quad N(x,y) = x^n N_1\left(\frac{y}{x}\right)$$

OR, $$M(x,y) = y^n M_1\left(\frac{x}{y}\right), \quad N(x,y) = y^n N_1\left(\frac{x}{y}\right)$$

Problem 1: Solve $\frac{dy}{dx} = \frac{y}{x} + \tan\frac{y}{x}$

Solution: Given that, $\frac{dy}{dx} = \frac{y}{x} + \tan\frac{y}{x}$ --- (1)

Let, $y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

From (1) we get,

$$v + x\frac{dv}{dx} = v + \tan v$$

$$\Rightarrow x\frac{dv}{dx} = \tan v$$

$$\Rightarrow \frac{dv}{\tan v} = \frac{dx}{x}$$

$$\Rightarrow \int \cot v\, dv = \int \frac{dx}{x} \quad \text{[By integrating]}$$

$$\Rightarrow \ln \sin v = \ln x + \ln c$$

$$\Rightarrow \ln \sin v = \ln xc$$

$$\Rightarrow \sin v = xc$$

$$\Rightarrow \sin\left(\frac{y}{x}\right) = cx$$

which is the required solution.

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Problem 2: Solve $(x^2+y^2)\,dx = 2xy\,dy$.

Solution: Given that, $(x^2+y^2)\,dx = 2xy\,dy$

$\Rightarrow \frac{dy}{dx} = \frac{x^2+y^2}{2xy} \quad \cdots (1)$

let $y = vx$ $\therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$

from (1) $\Rightarrow$ $v + x\frac{dv}{dx} = \frac{x^2+v^2x^2}{2x^2v}$

$\Rightarrow v + x\frac{dv}{dx} = \frac{1+v^2}{2v}$

$\Rightarrow x\frac{dv}{dx} = \frac{1+v^2}{2v} - v$

$\Rightarrow x\frac{dv}{dx} = \frac{1+v^2-2v^2}{2v}$

$\Rightarrow x\frac{dv}{dx} = \frac{1-v^2}{2v}$

$\Rightarrow \frac{2v}{1-v^2}\,dv = \frac{dx}{x}$

$\Rightarrow -\left(\frac{2v}{v^2-1}\right) - \frac{dx}{x} = 0$

$\Rightarrow \int\frac{2v}{v^2-1} + \int\frac{dx}{x} = 0$ [By integrating]

$\Rightarrow \ln(v^2-1) + \ln x = \ln c$

$\Rightarrow \ln x(v^2-1) = \ln c$

$\Rightarrow x(v^2-1) = c$

$\Rightarrow x\left(\frac{y^2}{x^2}-1\right) = c$ $\left[\because v = \frac{y}{x}\right]$

$\Rightarrow y^2 - x^2 = cx$

which is the required solution.

Problem 3: Solve $(x^2+y^2)dx = xy\,dy$

Solution: Given that, $(x^2+y^2)dx = xy\,dy$

$$\frac{dy}{dx} = \frac{x^2+y^2}{xy} \quad \text{---}(1)$$

Let, $y = vx \quad \therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$

From (1) $\Rightarrow v + x\frac{dv}{dx} = \frac{x^2+v^2x^2}{x^2v}$

$\Rightarrow v + x\frac{dv}{dx} = \frac{1+v^2}{v}$

$\Rightarrow x\frac{dv}{dx} = \frac{1+v^2}{v} - v$

$\Rightarrow x\frac{dv}{dx} = \frac{1}{v}$

$\Rightarrow v\,dv = \frac{dx}{x}$

$\Rightarrow$ By integrating

$\int v\,dv = \int \frac{dx}{x}$

$\Rightarrow \frac{v^2}{2} = \ln x - \ln c$

$\Rightarrow \ln x = \frac{v^2}{2} + \ln c$

$\Rightarrow \ln x = \frac{y^2}{2x^2} + \ln c$

$\Rightarrow x = ce^{\frac{y^2}{2x^2}}$

which is the required solution.

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Problem 4: Solve $(x+y)dy+(x-y)dx=0$

Solution: Given that, $(x+y)dy+(x-y)dx=0$

$$\Rightarrow (x+y)dy=-(x-y)dx$$

$$\Rightarrow \frac{dy}{dx}=-\frac{(x-y)}{x+y}$$

$$\Rightarrow \frac{dy}{dx}=\frac{y-x}{x+y} \quad \text{--- (1)}$$

Let, $y=vx \quad \therefore \frac{dy}{dx}=v+x\frac{dv}{dx}$

From (1) we get,

$$v+x\frac{dv}{dx}=\frac{vx-x}{x+vx}$$

$$\Rightarrow v+x\frac{dv}{dx}=\frac{x(v-1)}{x(v+1)}$$

$$\Rightarrow v+x\frac{dv}{dx}=\frac{v-1}{v+1}$$

$$\Rightarrow x\frac{dv}{dx}=\frac{v-1}{v+1}-v$$

$$\Rightarrow x\frac{dv}{dx}=\frac{v-1-v^2-v}{v+1}$$

$$\Rightarrow x\frac{dv}{dx}=-\frac{v^2+1}{v+1}$$

$$\Rightarrow \frac{v+1}{v^2+1}dv=-\frac{dx}{x}$$

$$\Rightarrow \frac{v+1}{v^2+1}dv+\frac{dx}{x}=0$$

$$\Rightarrow \frac{v}{v^2+1}dv+\frac{1}{v^2+1}dv+\frac{dx}{x}=0$$

$$\Rightarrow \frac{1}{2}\int\frac{2v}{v^2+1}dv+\int\frac{1}{v^2+1}dv+\int\frac{dx}{x}=0 \quad \text{[By integrating]}$$

$$=\frac{1}{2}\ln(v^2+1)+\tan^{-1}v+\ln x=c$$

$$\Rightarrow \frac{1}{2}\ln\left(\frac{y^2}{x^2}+1\right)+\tan^{-1}\left(\frac{y}{x}\right)+\ln x=c$$

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$\Rightarrow \tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{2}\ln\left(\frac{y^2+x^2}{x^2}\right) x = c$

$= \tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{2}\ln\left(\frac{y^2+x^2}{x}\right) = c$

which is the required solution.

Problem 5: Solve $2\frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{x^2}$

Solution: Given that, $2\frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{x^2}$

$\Rightarrow \frac{dy}{dx} = \frac{xy+y^2}{2x^2}$ --- (1)

Let $y = vx \therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$

From (1) we get,

$v + x\frac{dv}{dx} = \frac{x^2v + x^2v^2}{2x^2}$

$\Rightarrow v + x\frac{dv}{dx} = \frac{v+v^2}{2}$

$\Rightarrow x\frac{dv}{dx} = \frac{v+v^2}{2} - v$

$\Rightarrow x\frac{dv}{dx} = \frac{v+v^2-2v}{2}$

$\Rightarrow x\frac{dv}{dx} = \frac{v^2-v}{2}$

$\Rightarrow \frac{2dv}{v^2-v} = \frac{dx}{x}$

$\Rightarrow \frac{2dv}{v(v-1)} = \frac{dx}{x}$

$\Rightarrow 2\int\left(\frac{1}{v-1} - \frac{1}{v}\right)dv = \int\frac{dx}{x}$ By integrating

$\Rightarrow 2\ln(v-1) - 2\ln v = \ln x + \ln c$

$\Rightarrow 2\ln\frac{v-1}{v} = \ln cx$

$\Rightarrow \ln\left(\frac{v-1}{v}\right)^2 = \ln cx$

$$\left(\frac{v-1}{v}\right)^2 = cx$$

$$\Rightarrow (v-1)^2 = cxv^2$$

$$\Rightarrow \left(\frac{y}{x}-1\right)^2 = cx.\frac{y^2}{x^2} \quad \left[\because v=\frac{y}{x}\right]$$

$\Rightarrow (y-1)^2 = cxy^2$ which is the required solution.

~~Exact differential equation:~~

Problem 6: Solve $(x^3+3xy^2)dx + (y^3+3x^2y)dy = 0$

Solution: Given that,

$$(x^3+3xy^2)dx + (y^3+3x^2y)dy = 0$$

$$\Rightarrow \frac{dx}{dy} = -\frac{x^3+3xy^2}{y^3+3x^2y} \quad \cdots (1)$$

let $y = vx \quad \therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$

From (1) we get,

$$v + x\frac{dv}{dx} = -\frac{x^3+3x.x^2v^2}{x^3v^3+3x^2.vx}$$

$$\Rightarrow v + x\frac{dv}{dx} = -\frac{x^3(1+3v^2)}{x^3(v^3+3v)}$$

$$\Rightarrow v + x\frac{dv}{dx} = -\frac{1+3v^2}{v^3+3v}$$

$$\Rightarrow x\frac{dv}{dx} = -\frac{1+3v^2}{v^3+3v} - v$$

$$\Rightarrow x\frac{dv}{dx} = -\frac{1-3v^2-v^4-3v^2}{v^3+3v}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{-1-6v^2-v^4}{v^3+3v}$$

$$\Rightarrow \frac{\cancel{v^3+3v}}{1\theta}\, x\frac{dv}{dx} = \frac{-(1+6v^2+v^4)}{v^3+3v}$$

$$\Rightarrow \cancel{x}\frac{dv}{\cancel{dx}}\ \frac{v^3+3v}{1+6v^2+v^4}\,dv = -\frac{dx}{x}$$

$$\Rightarrow \frac{4v^3+12v}{1+6v^2+v^4}\,dv = -\frac{4dx}{x}$$

$$\Rightarrow \frac{4v^3+12v}{v^4+6v^2+1}\,dv + \frac{4dx}{x} = 0$$

By integrating we get,

$$\ln(v^4+6v^2+1) + 4\ln x = \ln c$$

$$\Rightarrow \ln(v^4+6v^2+1) + \ln x^4 = \ln c$$

$$\Rightarrow \ln(v^4+6v^2+1)x^4 = \ln c$$

$$\Rightarrow (v^4+6v^2+1)x^4 = c$$

$$\Rightarrow \left(\frac{y^4}{x^4}+\frac{6y^2}{x^2}+1\right)x^4 = c$$

$$\Rightarrow y^4+6x^2y^2+x^4 = c$$

which is the required solution.

Exact differential equation: A differential equation $M(x,y)\,dx + N(x,y)\,dy = 0$ is called exact equation if $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$.

Solution of exact: $\int M(x,y)\,dx + \int N(y)\,dy = 0$

x বর্জিত পদ

Problem 1: Solve $(2x+3y-6)\,dy = (6x-2y-7)\,dx$

Solution: Given that, $(2x+3y-6)\,dy = (6x-2y-7)\,dx$

~~Here, M = 6x-2y-7, N = 2x+3y-6~~

$$\Rightarrow (6x-2y-7)\,dx - (2x+3y-6)\,dy = 0$$

$$\Rightarrow (6x-2y-7)\,dx + (-2x$$

Exact differential equation: A differential equation $M(x,y)dx + N(x,y)dy = 0$ is called exact equation if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Solution of exact: $\int M(x,y)\,dx + \int\limits_{x \text{ বর্জিত পদ}} N(y)\,dy = 0$

Question 01: Solve $(3x^2y - 6x)dx + (x^3 + 2y)dy = 0$

Solution: Given that $(3x^2y - 6x)dx + (x^3+2y)dy = 0$

Here, $M = 3x^2y - 6x$, $N = x^3 + 2y$

$$\frac{\partial M}{\partial y} = 3x^2 \qquad \frac{\partial N}{\partial x} = 3x^2$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so the DE is exact.

The solution is: $\int M(x,y)\,dx + \int N(y)\,dy = 0$

$$\cancel{\int x^3+3xy} \int (3x^2y - 6x)dx + \int 2y\,dy = 0$$

$$\Rightarrow 3\frac{x^3}{3}y - 6\frac{x^2}{2} + 2\frac{y^2}{2} = c$$

$$\Rightarrow x^3y - 3x^2 + y^2 = c$$ Ans

Question 02 Solve $(6x-2y-7)dx+(6-2x-3y)dy=0$

Solution: Given that, $(6x-2y-7)dx + (6-2x-3y)dy = 0$

Here, $M = 6x-2y-7$, $N = 6-2x-3y$

$\frac{\partial M}{\partial y} = -2$ $\frac{\partial N}{\partial x} = -2$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ so the DE is exact.

The solution is: $\int M(x,y)dx + \int N(y)dy = 0$

$\Rightarrow \int(6x-2y-7)dx + \int(6-3y)dy = 0$

$\Rightarrow 3x^2 - 2xy - 7x + 6y - \frac{3}{2}y^2 = c$

$\Rightarrow 6x^2 - 4xy - 14x + 12y - 3y^2 = c'$ Ans

Question-03 Solve $(2x+y-1)dy + (2y-x-5)dx = 0$

Soln Here, $M = 2y-x-5$ $N = 2x+y-1$

$\frac{\partial M}{\partial y} = 2$ $\frac{\partial N}{\partial x} = 2$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, So the DE is exact

The soln is: $\int M(x,y)dx + \int N(y)dy = 0$

$\Rightarrow \int(2y-x-5)dx + \int(y-1)dy = 0$

$\Rightarrow 2xy - \frac{x^2}{2} - 5x + \frac{y^2}{2} - y = c$

$\Rightarrow 4xy - x^2 - 10x + y^2 - 2y = c'$

$\Rightarrow y^2 + 4xy - x^2 - 10x - 2y = c'$ Ans

Question 04: Solve $(x^2 - ay)dx + (y^2 - ax)dy = 0$

Soln: Given that, $(x^2 - ay)dx + (y^2 - ax)dy = 0$

Here, $M = x^2 - ay$, $N = y^2 - ax$

$\frac{\partial M}{\partial y} = -a$ $\qquad \frac{\partial N}{\partial x} = -a$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ So the DE is exact

The solution is: $\int M(x,y)dx + \int N(y)dy = 0$

$\Rightarrow \int (x^2 - ay)\,dx + \int y^2 dy = 0$

$\Rightarrow \frac{x^3}{3} - axy + \frac{y^3}{3} = c$

$\Rightarrow x^3 - 3axy + y^3 = c'$ Ans.

Solve $(x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 0$

Soln: Given that, $(x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 0$

Here, $M = x^3 + 3xy^2$ $\qquad N = y^3 + 3x^2y$

$\frac{\partial M}{\partial y} = 6x$ $\qquad \frac{\partial N}{\partial x} = 6xy$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$ So the DE is exact

The solution is: $\int M(x,y)dx + \int N(y)dy = 0$

$\Rightarrow \int (x^3 + 3xy^2)dx + \int y^3 dy = 0$

$\Rightarrow \frac{x^4}{4} + \frac{3}{2}x^2y^2 + \frac{y^4}{4} = c$

$\Rightarrow x^4 + 6x^2y^2 + y^4 = c'$

Ans.

Ex solve $(\tan y - 3x^2)dx + x\sec^2 y\, dy = 0$

Soln: Given that, $(\tan y - 3x^2)dx + x\sec^2 y\, dy = 0$

Here, $M = \tan y - 3x^2$ $\quad N = x\sec^2 y$

$\frac{\partial M}{\partial y} = \sec^2 y$ $\quad \frac{\partial N}{\partial x} = \sec^2 y$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. So the DE is exact.

~~The differential eqn is:~~

The solution is: $\int M(x,y)\,dx + \int N(y)\,dy = 0$

$\Rightarrow \int(\tan y - 3x^2)dx + \int 0\, dy = 0$

$\Rightarrow x\tan y - x^3 + 0 = c$

$\Rightarrow x\tan y - x^3 = c$ Ans

Ex solve $(e^x \sin y + e^{-y})dx + (e^x \cos y - xe^{-y})dy = 0$

Soln. Here, $M = e^x \sin y + e^{-y}$, $N = e^x \cos y - xe^{-y}$

$\frac{\partial M}{\partial y} = e^x \cos y - e^{-y}$ $\quad \frac{\partial N}{\partial x} = e^x \cos y - e^{-y}$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. So the DE is exact

The solution is: $\int M(x,y)\,dx + \int N(y)\,dy = 0$

$\Rightarrow \int(e^x \sin y + e^{-y})dx + N\int 0\, dy = 0$

$\Rightarrow e^x \sin y + xe^{-y} = c$ A